

MATH 3430 - Week 2

Second-Order Equations and Planar Systems

1 Overview

There are two main topics this week:

1. Second-Order Differential Equations
2. Systems of First-Order Differential Equations

It turns out that these topics are very closely related, as we will see.

We will start with equations of the form $a(t)x'' + b(t)x' + c(t)x = f(t)$. When $f(t) = 0$ the equation is called **homogeneous**. When $f(t) \neq 0$, then it is **nonhomogeneous**. If $a(t) = a, b(t) = b, c(t) = c$ are all constants, then the equation is called **constant-coefficient**. We will spend most of our time this week working with constant-coefficient homogeneous equations, where the theory is essentially complete.

We will then move to systems of first-order differential equations of the form

$$\begin{aligned}x' &= f(x, y) \\ y' &= g(x, y)\end{aligned}$$

The simplest case is when both $f(x, y)$ and $g(x, y)$ are linear functions: $f(x, y) = ax + by$ and $g(x, y) = cx + dy$. These are called **Planar Systems**. We will use tools from linear algebra to study these planar systems and will investigate them from a geometric (phase plane, vector fields, equilibrium points, stability) as well as an analytical (explicit solutions in terms of the eigenvalues and eigenvectors of the **coefficient matrix**) perspective.

While we will focus a lot on linear equations and systems, we will also look at some important nonlinear systems. We return to nonlinear systems in Week 4, so the material this week will be more of a quick introduction instead of a systematic treatment. We will look at interacting species models and will focus on the qualitative side of things. We will rely heavily on Sage to help of with all of this.

Finally, we will return to second-order differential equations and look at how we can use power-series to find **series solutions** to certain important and famous second-order equations, focusing mostly on Bessel's Equation and the related Bessel Functions of the First Kind.

2 Schedule and Checklist

Here is a brief schedule for the week. Please note that this is subject to change, so you need to adjust to what we do in class, regardless of what it says here. Think of this as a pretty accurate guideline, though.

- (M) Bifurcations and Bifurcation Diagrams. Harmonic Oscillators. Second-Order Constant Coefficient Equations. The Characteristic Polynomial. General Solutions. The Wronskian.

- (T) 2x2 Linear Systems (Planar Systems). Eigenvalues and Eigenvectors. General Solutions.
- (W) Direction Fields. Equilibrium Points and Stability. The Trace-Determinant Plane.
- (R) Interacting Species. Nonlinear 2x2 Systems. Nullclines.
- (F) Power Series Methods. Bessel's Equation and Bessel Functions.

Here is a checklist of things to do this week. This is meant to help you keep track of what you've completed and what you still need to work on.

- ☐ Read these notes and ask any questions that you have about anything.
- ☐ Read the sections from Judson and Lebl.
- ☐ Complete all of the homework exercises.

3 Terminology and Notation

- Bifurcations and Bifurcation Diagrams
- Second-Order Linear Equations
 - Constant Coefficient
 - Homogeneous
 - Nonhomogeneous/Forced
- Characteristic Polynomial
- Second-Order Nonlinear Equations
- Planar Systems (2x2 Linear Systems)
- Identity Matrix
- Trace and Determinant of a 2x2 Matrix
- Eigenvalues and Eigenvectors
- Complex Numbers and Euler's Formula
- Vector Fields
- Equilibrium Points
 - Sink
 - Source
 - Saddle
 - Spiral Sink
 - Spiral Source
 - Center
- Trace-Determinant Plane
- Nullclines
- Nonlinear Systems
- Power Series Solutions

4 Equations and Models

- Harmonic Oscillators

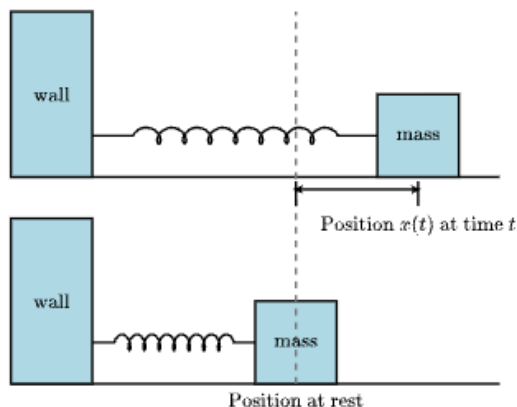


Figure 1: Spring-Mass System (Harmonic Oscillator)

Let an object of mass m be attached to a fixed object (such as a wall) by a spring. Let $x(t)$ denote the displacement of the object from its equilibrium position at time t . Then, using Newton's Second Law of Motion and Hooke's Law, the function $x(t)$ satisfies the second-order differential equation

$$mx'' + bx' + kx = f(t)$$

where $m > 0$ is the mass of the object, $b \geq 0$ is the **damping coefficient** (think friction), $k > 0$ is the **spring constant** determined by material properties of the spring, and $f(t)$ is the **driving force** on the system. If $b = 0$ then the system is said to be **undamped**. If $f(t) = 0$, then the system is said to be **unforced**.

- Electrical Circuits

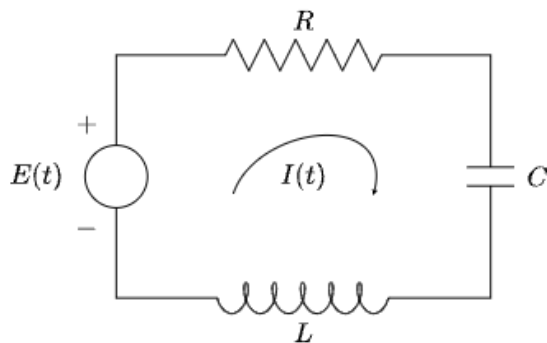


Figure 2: A simple RLC circuit

The diagram shows the schematic for a simple **RLC Circuit**. In any circuit, the main objects under consideration are the **charge** on the capacitor at time t , which is denoted by $Q(t)$ and the **current** through the capacitor, which is denoted by $I(t)$. The relationship between the two is given by $I(t) = Q'(t)$. The function $E(t)$ is the voltage (from a battery or some other external source), and

the constants, R, C, L are determined by the components of the circuit. R is the resistance, C is the capacitance, and L is the inductance.

Using various physical laws (Faraday's Law and Kirchoff's Laws) we have the following equivalent second-order differential equations

$$LQ'' + RQ' + \frac{1}{C}Q = E(t), \quad \text{and} \quad LI'' + RI' + \frac{1}{C}I = E'(t)$$

- Interacting Species

There is a large family of models for the interaction of two species. Let $x(t)$ be the population of species 1 at time t , and let $y(t)$ be the population of species 2 at time t . Then the general form of the system of differential equations modeling the interaction of the two species is given by:

$$\begin{aligned} \frac{dx}{dt} &= \alpha f_1(x) + \beta xy + h_1(t) \\ \frac{dy}{dt} &= \gamma f_2(y) + \delta xy + h_2(t) \end{aligned}$$

The parameters have the following meaning:

- α : growth rate for species 1 (how fast species 1 grows/decays without any interaction with species 2)
- β : interaction parameter between species 1 and species 2 (how the interaction of species 1 with species 2 influences the growth of species 1)
- γ : growth rate for species 2 (how fast species 2 grows/decays without any interaction with species 1)
- δ : interaction parameter between species 2 and species 1 (how the interaction of species 2 with species 1 influences the growth of species 2)

The functions f are usually either

$$\begin{aligned} f_1(x) &= x & f_2(y) &= y & & \text{(exponential growth)} \\ f_1(x) &= x \left(1 - \frac{x}{M}\right) & f_2(y) &= y \left(1 - \frac{y}{N}\right) & & \text{(logistic growth)} \end{aligned}$$

Here, M, N are the carrying capacity for the two species.

The functions h are usually equal to a constant and denote harvesting terms. In most cases there will be no harvesting and so $h = 0$.

There are three main situations that we will look at are determined by the signs of β and δ , and in one case also γ :

- Competing Species: $\beta < 0$ and $\delta < 0$. The interaction between species 1 and 2 decreases the overall population growth of both species. This models the situation where the species are competing with one another.

- Cooperating Species: $\beta > 0$ and $\delta > 0$. The interaction between species 1 and 2 increases the overall population growth of both species. This models the situation where the species are cooperating with one another.
- Predator-Prey Systems: $\beta < 0$, $\gamma < 0$, and $\delta > 0$. The interaction of species 1 (the prey) with species 2 (the predator) decreases the growth of species 1 but increases the growth of species 2. This models the situation where species 2 preys upon species 1. **PLEASE NOTE:** in this situation, the growth rate for species 2 is negative. Species 2 needs to feed upon species 1 in order to survive. If there is no interaction between the two species, then species 2 will starve and the population will decrease. That's what the $\gamma < 0$ signifies. If $\alpha < 0$, $\beta > 0$, and $\delta < 0$, then species 1 preys upon species 2.

- Airy Functions

The **Airy Equation** is given by

$$y'' - xy = 0$$

Using power series, you can show that

$$y = a_0 \left(1 + \frac{1}{6}x^3 + \frac{1}{180}x^6 + \cdots + \frac{1}{2 \cdot 3 \cdot 5 \cdot 6 \cdots (3n-1) \cdot (3n)}x^{3n} + \cdots \right) \\ + a_1 \left(x + \frac{1}{12}x^4 + \frac{1}{504}x^7 + \cdots + \frac{1}{3 \cdot 4 \cdot 6 \cdot 7 \cdots (3n) \cdot (3n+1)}x^{3n+1} + \cdots \right)$$

Let

$$y_1(x) = 1 + \frac{1}{6}x^3 + \frac{1}{180}x^6 + \cdots + \frac{1}{2 \cdot 3 \cdot 5 \cdot 6 \cdots (3n-1) \cdot (3n)}x^{3n} + \cdots \\ y_2(x) = x + \frac{1}{12}x^4 + \frac{1}{504}x^7 + \cdots + \frac{1}{3 \cdot 4 \cdot 6 \cdot 7 \cdots (3n) \cdot (3n+1)}x^{3n+1} + \cdots$$

Then the general solution (which will depend on the initial conditions $y(0) = a_0$ and $y'(0) = a_1$) is given by

$$y(x) = a_0 y_1(x) + a_1 y_2(x)$$

The graphs of $y_1(x)$ and $y_2(x)$ are shown below (these correspond to the initial conditions $a_0 = 1, a_1 = 0$ and $a_0 = 0, a_1 = 1$)

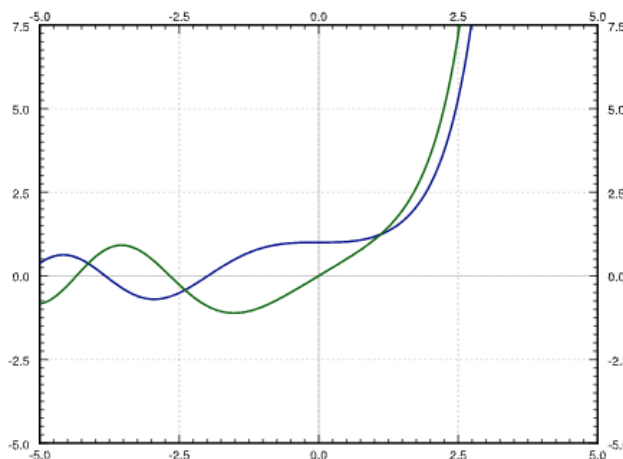


Figure 3: The solutions $y_1(x)$ and $y_2(x)$.

Please note that these solutions have oscillatory behavior for $x < 0$ and exponential behavior for $x > 0$. If you look at the differential equation, this shouldn't be surprising (why?).

- Bessel's Differential Equation and the Bessel Functions $J_0(x)$ and $J_1(x)$

The **Bessel Equation** is given by

$$x^2 y'' + xy' + (x^2 - p^2)y = 0$$

The most important case for us is when p is a nonnegative integer, n .

Using power series, we can show that a solution, denoted $J_n(x)$, is given by

$$J_n(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(k+n)!} \left(\frac{x}{2}\right)^{2k+n}$$

These functions are called **Bessel Functions of the first kind**. There are related linearly independent solutions called **Bessel Functions of the second kind**. They are denoted by $Y_n(x)$. We won't explicitly write down their definition. The general solution (where the constants are determined by the initial conditions) is then given by

$$y(x) = AJ_n(x) + BY_n(x)$$

The most commonly occurring Bessel Functions are $J_0(x)$ and $J_1(x)$. In these cases we have

$$J_0(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2} \left(\frac{x}{2}\right)^{2k} = \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{2k}(k!)^2} x^{2k}$$

$$J_1(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(k+1)!} \left(\frac{x}{2}\right)^{2k+1} = \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{2k+1}k!(k+1)!} x^{2k+1}$$

The graphs of $J_0(x)$ and $J_1(x)$ are shown below

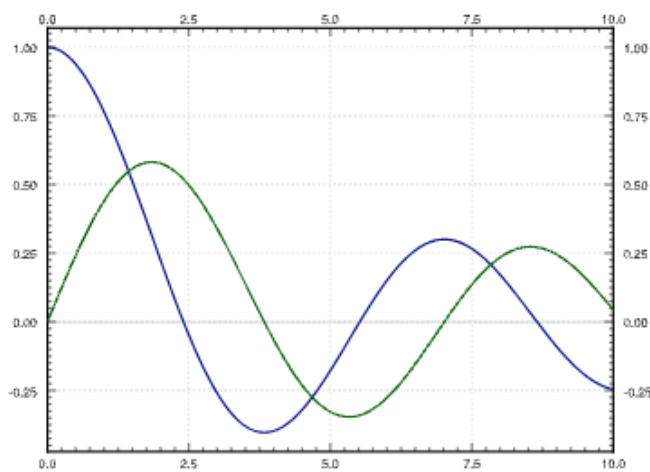


Figure 4: The solutions $J_0(x)$ and $J_1(x)$.

5 Important Formulas (analytical material)

- (Existence and Uniqueness Theorem for Second-Order Linear Differential Equations) Suppose that p, q, f are continuous functions on some interval I , that x_0 is some number in I , and that b_0, b_1 are any constants. Then the equation

$$y'' + p(x)y' + q(x)y = f(x), \quad y(x_0) = b_0, \quad y'(x_0) = b_1$$

has exactly one solution $y(x)$ defined on the interval I .

- The trace of a 2x2 matrix is

$$\text{tr} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = a + d$$

- The determinant of a 2x2 matrix is

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

- To find the eigenvalues of a matrix A , solve the following equation for λ :

$$\det(A - \lambda I) = 0$$

- Euler's formula says that

$$e^{i\theta} = \cos(\theta) + i \sin(\theta)$$

- Given the complex number $z = \alpha + i\beta$, Euler's formula says that

$$e^z = e^{\alpha+i\beta} = e^\alpha e^{i\beta} = e^\alpha (\cos(\beta) + i \sin(\beta))$$

- If the 2x2 matrix A has two distinct real eigenvalues λ_1 and λ_2 (with corresponding eigenvectors v_1 and v_2), then the general solution to the differential equation $\mathbf{x}' = A\mathbf{x}$ is given by

$$x(t) = c_1 e^{\lambda_1 t} v_1 + c_2 e^{\lambda_2 t} v_2$$

- Let the 2x2 real-valued matrix A have complex conjugate eigenvalues $\lambda = \alpha \pm i\beta$. If v is the (complex-valued) eigenvector corresponding to the eigenvalue $\alpha + i\beta$ (everything would also work out if you let v be the eigenvector for the eigenvalue $\alpha - i\beta$). Then the system $x' = Ax$ has two linearly independent solutions

$$x_1(t) = \text{Re} \left(e^{(\alpha+i\beta)t} v \right), \quad x_2(t) = \text{Im} \left(e^{(\alpha+i\beta)t} v \right)$$

where $\text{Re}(z(t))$ is the **real part** of the complex function $z(t)$ and $\text{Im}(z(t))$ is the **imaginary part** of the complex function $z(t)$.

The general solution is then given by

$$x(t) = c_1 x_1(t) + c_2 x_2(t)$$

- Let the 2×2 real-valued matrix A have a single (real) eigenvalue λ . If v_1 and v_2 are linearly independent eigenvectors corresponding to λ , then the general solution to $x' = Ax$ is given by

$$x(t) = c_1 e^{\lambda t} v_1 + c_2 e^{\lambda t} v_2$$

If λ has only one linearly independent eigenvector v_1 , let v_2 be any vector that solves

$$(A - \lambda I)v_2 = v_1$$

Then $x' = Ax$ has two linearly independent solutions

$$x_1(t) = e^{\lambda t} v_1, \quad x_2(t) = e^{\lambda t} (v_2 + t v_1)$$

The general solution to $x' = Ax$ is given by

$$x(t) = c_1 x_1(t) + c_2 x_2(t)$$

PLEASE NOTE: the general solutions, $x(t)$, in all three cases listed above are 2×1 **vectors**. This gives two solutions, one from the first component and another from the second component, of $x(t)$. We will go over examples in class (and you can find more examples in Judson and Lebl), but please make sure you understand what this means in terms of solving the 2×2 system $x' = Ax$.

6 Geometric and Visual Aspects (qualitative material)

- Here is an example of a bifurcation diagram for the one-parameter family of differential equations connected to the logistic growth with harvesting model: $P' = -P^2 + 200P - 200H$. The horizontal axis is H and the vertical axis is P . There are no equilibrium solutions for $H > 50$. Any vertical line gives the phase line for that particular value of H .

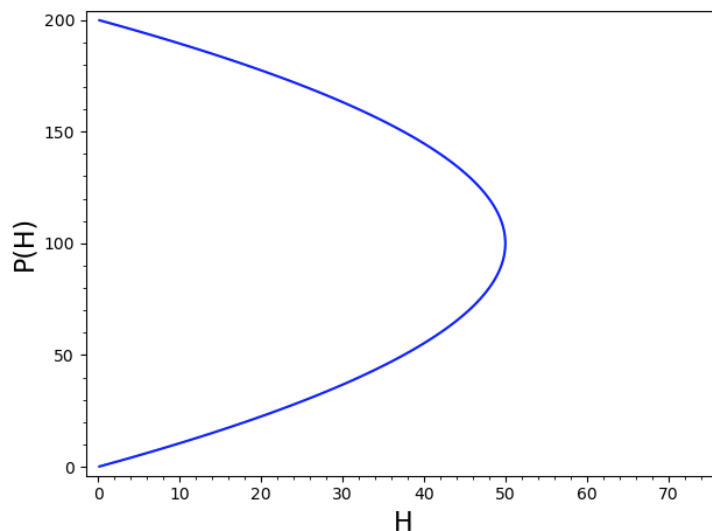


Figure 5: Bifurcation Diagram for $P' = -P^2 + 200P - 200H$

- The following table summarizes the nature of equilibrium points in terms of the eigenvalues of the matrix A :

Summary of behavior of linear homogeneous two-dimensional systems.

Eigenvalues	Behavior
real and both positive	source / unstable node
real and both negative	sink / stable node
real and opposite signs	saddle
purely imaginary	center point / ellipses
complex with positive real part	spiral source
complex with negative real part	spiral sink

- Trace-Determinant Plane in terms of eigenvalues of A :

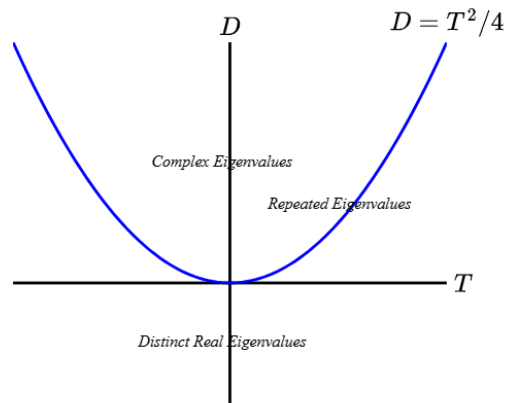


Figure 6: The Trace-Determinant Plane: Eigenvalues

- Trace-Determinant Plane in terms of equilibrium solutions:

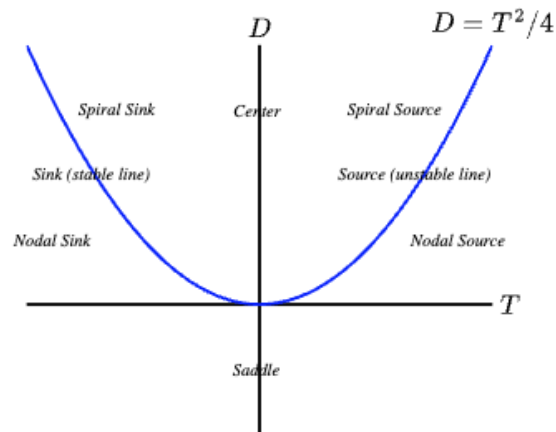


Figure 7: The Trace-Determinant Plane: Equilibrium Points

7 Sage Code (numerical material)

There is a lot of very useful Sage material in Chapter 2 of Judson. Specifically, section 2.1.7 and 2.2.9 are completely devoted to Sage and you should definitely take a look at them. There is also a ton of Sage-related material in section 2.4. Check it out!

7.1 Commands

- `implicit_plot(function==0, (horizontal variable, min, max), (vertical variable, min, max))`

Implicitly plots $f(x, y) = 0$. All of the standard plot options (like *thickness*, *color*, *labels*, etc.) can be used with this. You can change the right-hand side of the *function* == 0 to any constant (instead of 0).

- `matrix(m,n,[a11,a12,...,amn])`

This creates a $m \times n$ matrix with entries a_{ij} with $1 \leq i \leq m$ and $1 \leq j \leq n$. We will be working mainly with 2×2 matrices, which are given the following command.

- `matrix(2,2,[a,b,c,d])`

- `matrix([[a11,a12,...,a1n],[a21,a22,...,a2n], ..., [am1,am2,...,amn]])`

This is an alternate way of creating a matrix by entering the columns of the matrix as lists. It is sometimes easier to keep track of things if they're grouped together by columns. For 2×2 matrices this can be written as

- `matrix([[a,b],[c,d]])`

- `.eigenvalues()`

This gives a list of all the eigenvalues of the matrix that the command is appended to.

- `.right_eigenvectors()`

This gives a list of the (right) eigenvectors of the matrix the command is appended to. The list contains the eigenvalue, a basis for the eigenspace associated to the eigenvalue (these are linearly independent eigenvectors corresponding to the eigenvalue), and the algebraic multiplicity of the eigenvalue.

- `desolve_system([de1, de2], [x, y], ics=[t0,x(t0),y(t0)])`

This will try to find analytical solutions to the system of differential equations given by

$$\begin{aligned}x' &= f(x, y) \\ y' &= g(x, y)\end{aligned}$$

The functions *de1*, *de2* are defined by `de1= diff(x,t)-f(x,y)==0` and `de2 = diff(y,t)-g(x,y)==0` must be defined earlier in the code. If you are trying to solve an IVP, use the `ics=[t0,x(t0),t(t0)]` option. If this option is omitted, Sage will try to find the general solution to the system.

- `desolve_system_rk4([f,g], [x,y], ivar=t, ics=[t0,x(t0),y(t0)],end_points=t_end, step=step_size)`

This code numerically finds solution curves that start at $(x(t_0), y(t_0))$. The algorithm starts with the value $t = t_0$ and continues until $t = t_end$, taking steps of size $\Delta t = step_size$. The output of this command is a huge list of points of the form $(t_i, x(t_i), y(t_i))$.

7.2 Example Code

- Plotting the implicit function $-P^2 + 200P - 200H = 0$. This gives a bifurcation diagram for the one-parameter family of functions $f_H(P) = -P^2 + 200P - 200H$. Note that this function is an expanded form of the function on the right-hand side of Logistic Growth with Harvesting equation $P' = P(1 - P/200) - H$. This code, then, plots the **bifurcation diagram** for this family of equations, depending on the harvesting parameter H .

```
H = var('H')
P = function('P')(H)
f(H,P) = -P^2+200*P-200*H
implicit_plot(f==0, (H, 0, 75), (P, 0, 200), axes_labels=['H', 'P(H)'])
```

- This code constructs a 2×2 matrix A and shows some of the linear algebra commands that you can use on A :

```
A = matrix(2,2,[1,2,-1,4])
show(A)
show(A.eigenvalues())
show(A.right_eigenvectors())
show(A.det())
show(A.charpoly())
```

- This code uses the `desolve_system()` command to find explicit solutions to an IVP:

```
a , b , c, d = 0, 1, -5, -4
x0, y0 = 1, 0
t = var('t')
x = function('x')(t)
y = function('y')(t)
de1 = diff(x,t) - a*x - b*y == 0
de2 = diff(y,t) - c*x - d*y == 0
sol = desolve_system([de1, de2], [x, y], ivar = t, ics=[0, x0, y0])
sol
```

- This code plots the solution curves to a 2×2 system that can be explicitly solved using the `desolve_system()` command:

```
a , b , c, d = 0, 1, -5, -4
x0, y0 = 1, 0
t = var('t')
x = function('x')(t)
y = function('y')(t)
sol1 = function('sol1')(t)
sol2 = function('sol2')(t)
```

```

de1 = diff(x,t) - a*x - b*y == 0
de2 = diff(y,t) - c*x - d*y == 0
sol = desolve_system([de1, de2], [x, y], ivar = t, ics=[0, x0, y0])
sol1 = sol[0].rhs()
sol2 = sol[1].rhs()
p = plot(sol1, t, 0, 5)
p += plot(sol2, t, 0, 5, color = 'red')
p.show()

```

- Generating and plotting the vector field and solution curve for the harmonic oscillator (this code can be modified to plot solution curves for any linear system by changing the values of a, b, c, d in the first line of the code). Note that this example uses the `desolve_system_rk4` command.

```

a , b , c, d = 0, 1, -9, 0
x0, y0 = 2, 1
x, y = var('x y')
F = [a*x + b*y, c*x + d*y]
n = sqrt(F[0]^2 + F[1]^2)
p = plot_vector_field((F[0]/n, F[1]/n), (x, -6, 6), (y, -6, 6), aspect_ratio = 1)
L = desolve_system_rk4(F, [x, y], ics=[0, x0, y0], ivar = t, endpoints = 10, step = 0.05)
S = [ [j, k] for i, j, k in L]
p += line(S, axes_labels=['x(t)', 'y(t)'], xmin = -6, xmax = 6, ymin = -6, ymax = 6)
p.show()

```

- Generating and plotting the solution curves for the harmonic oscillator:

```

b, c = 1, 1
t_start, x_start, y_start = 0, 0, 1
x, y, t = var('x y t')
F = [y, -c*x - b*y]
L = desolve_system_rk4(F, [x, y], ics = [t_start, x_start, y_start], ivar = t, endpoints
= 7, step = 0.1)
x_sol = [[i, j] for i,j,k in L]
y_sol = [[i, k] for i,j,k in L]
p = line(x_sol, color='red', legend_label='x(t)', legend_color='red')
p += line(y_sol, color='blue', legend_label='y(t)=x'(t)', legend_color='blue')
p.show()

```

- Generating and plotting the vector field and a solution curve in the phase plane for the predator-prey nonlinear system:

```

a , b , c, d = 2, -1, -5, 1
x, y, t = var('x y t')
F = [a*x + b*x*y, c*y + d*x*y]
n = sqrt(F[0]^2 + F[1]^2)
p = plot_vector_field((F[0]/n, F[1]/n), (x, 0, 15), (y, 0, 15), aspect_ratio = 1)
L = desolve_system_rk4(F, [x, y], ics=[0, 1, 1], ivar = t, endpoints = 10, step = 0.01)
S = [ [j, k] for i, j, k in L]
p += line(S, axes_labels=['R(t)', 'F(t)'], xmin = 0, xmax = 15, ymin = 0, ymax = 15)
p.show()

```

8 General Comments

- We've only been looking at 2×2 linear systems. There are a couple of reasons for this. First, they correspond to second-order differential equations, which are the most important and the most common. Second, they are 'small' enough that the geometry of what's going on can be visualized easily. We're looking at 2-dimensional vectors and the vector fields live in the plane. Solution curves also live in the plane, which makes them particularly easy to graph. Finally, on a more algebraic side, we can always find the eigenvalues of a 2×2 system by using the quadratic formula. This means that we can solve any 2×2 system of linear differential equations. Connected to this, the determinant of a 2×2 matrix is very easy to compute, which definitely is not the case for larger matrices.
- With all of that being said, basically everything that we've done goes through in a natural way to higher-order systems. We will spend some time during the last week looking at 3×3 systems. In this case, we can still work with things on a geometric level, but now the vectors are 3-dimensional and the vector fields live in space. Computers can graph these things for us, but it's not nearly as easy to actually 'see' what's going on. Once we move to 4×4 systems or higher we have to pretty much abandon the geometric side of things. We also run into serious problems when it comes to finding eigenvalues. While there are cubic and even quartic formulas, no one actually knows what they are or uses them. Once we get to 5×5 and above, there is no surefire way to even solve polynomial equations, so eigenvalues are out of the question. Not to mention that the determinant of bigger matrices are very difficult to work with. But, as I've already said, we will look at some higher-order systems and it's important to bear in mind that all of the conceptual things that we did in the 2×2 case go through the same way (in theory, at least).
- In many ways, this material is the heart of the course. It is a relatively 'modern' approach to the subject, replacing the more traditional analytic techniques that dominated the 18th and 19th century approach to the subject. As with most things modern, it relies very heavily on modern technology, namely computers. But, if we have the tools, we should use them, right?
- While on the topic of 'modern' versus 'traditional,' power series solutions to second-order differential equations is an example of a traditional analytic technique. It's a bit of an oversimplification, but one of the main differences between the two approaches is that traditional techniques are based far more on the tools and ideas of calculus, while the modern techniques are based much more on the tools and ideas of linear algebra. If you enjoy calculus and all things analysis, then you really should take a look at a more traditional book about differential equations (the one that comes immediately to mind is Whittaker and Watson, *A Course of Modern Analysis*). If you want to continue in this direction and learn more about Bessel functions, then you should find the book by Watson (the same Watson as before) called *A Treatise on the Theory of Bessel Functions*. It's 814 pages of all things Bessel.
- It was probably not very clear why such a big deal was made about eigenvalues in your Linear Algebra class. This week starts to give the answer. If you want to solve differential equations (and everyone wants to solve differential equations) then you need to be able to find eigenvalues. Once you have eigenvalues, you can build the solutions. It of course goes much deeper than this, but that really is the essence of it. There are equations that you want to solve. The solutions almost always involve the eigenvalues and eigenvectors of some related system. The deeper you get into mathematics the more you'll see this same pattern repeated, over and over again.

9 Reading Notes

- Bifurcations (J 1.7)
- Second-Order Equations (J 4.1) (L 2.2)
- Harmonic Oscillators (J 4.1.3) (L 2.4)
- Linear Algebra Review (J 3.1) (L 3.2, Appendix A)

- 2×2 Systems (J 3.2) (L 3.3, 3.5)
- Solutions using Eigenvalues and Eigenvectors (J 3.3, 3.4, 3.5) (L 3.4, 3.7)
- Vector Fields (J 3.3) (L 3.5)
- Phase Plane Analysis, Nullclines (J 2.2)
- Equilibrium Points and Stability (L 3.5)
- Trace-Determinant Plane (J 3.7)
- Interacting Species (J 2.1, 2.2)
- Power Series Solutions (L 7.2, 7.3)
- Bessel Functions (L 7.3.3)

10 Additional Topics

Look in the indices of Judson and Lebl for these topics.

- Undetermined Coefficients
- Variation of Parameters
- The Wronskian

11 Homework Exercises

Please do **ALL** the ‘To Turn In’ exercises. Not all of them will be graded for correctness, but they will all be checked for completion. Failure to do all of them will lower the completion portion of the Homework score.

Please write your solutions up clearly and cleanly, using full English sentences. Try not to have a bunch of arrows pointing from one thing to another or a bunch of things crossed-out and a bunch of other things circled, etc. Presentation matters.

11.1 To Turn In

- Pick one item from the Additional Topics and write a paragraph or two about it. This doesn’t have to be extensive, but it should give enough information to explain what the topic is, how it is used, what ideas are involved, etc.
- Judson 1.7) 6

This is a nice bifurcation diagram problem. The best thing to do is graph $\sin y$ and λy on the same set of axes (letting y be the independent variable) and think about where the graphs intersect for different values of λ . These points of intersection are where $\sin y = \lambda y$ which means that $\lambda y - \sin y = 0$. Desmos can be helpful here.

- Judson 2.1) 5
- Judson 2.2) 1, 3
- Judson 2.4) 2
- Judson 3.2) 6, 10

- Judson 3.3) 2, 7
- Judson 3.4) 2, 5, 6
- Judson 3.5) 10

For the exercises in sections 3.3-3.5, it's probably easier to use $\det(\lambda I - A) = 0$ instead of $\det(A - \lambda I) = 0$ (either one will work). There are less minus signs to keep track of this way, but it's up to you.

- Judson 3.7) 9, 16
- Lebl 7.2) 4

Okay, this one is a lot. You need to keep track of the indices of the sums and the powers of x . The easiest thing to do is probably write

$$y'' = 2a_2 + 6a_3 + \sum_{k=0}^{\infty} (k+4)(k+3)a_{k+4}x^{k+2} \quad \text{and} \quad x^2y = \sum_{k=0}^{\infty} a_kx^{k+2}$$

This will give you a recurrence relation between a_{k+4} and a_k . There will be four separate cases that you will need to consider: $k = 4n$ (which will be related to a_0), $k = 4n + 1$ (which will be related to a_1), $k = 4n + 2$ (which will be related to a_2), and $k = 4n + 3$ (which will be related to a_3). The good news is that two of these four situations will be very simple. The other two, not so much. For your answer, don't try to write it as a sum. Just write out the first 8 nonzero terms (this should take you up to the x^{13} term in the power series for y). Look at the 'For Practice' problem from section 7.2 in Lebl listed below for some help with this if you get stuck.

11.2 For Practice

- Lebl 2.2) 101, 102, 103, 104, 105, 106
- Lebl 3.1) 102
- Lebl 3.3) 104
- Lebl 3.4) 102, 103, 104
- Lebl 3.5) 101, 102, 103
- Lebl 7.2) 101